Mathematics: analysis and approaches
Standard level
Paper 2 — markscheme
4.1–4.6 Statistics and probability

13 October 2026

W4ss Academy practice set 2 | **Topic** 4.1–4.6 Statistics and probability | **Ref.** W4SS–MAA–T4–P2–02

Student name

Paper 2

Instructions to students

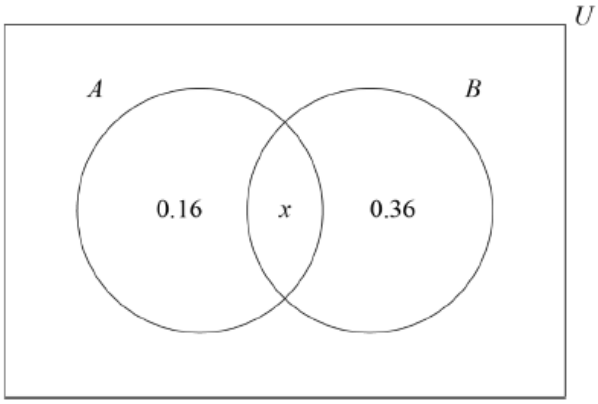
- This markscheme is for teachers and for self-assessment after the test.
- **M**: method marks. **A**: answer or accuracy marks, usually after the M mark they depend on. **R**: reasoning marks. **N** (former course): marks for a correct answer with no working.
- Marks in brackets, e.g. **(M1)**, can be implied by correct working that follows. **AG**: answer given in the question, no mark for it. **FT**: follow through from an earlier error.
- Total: **[26 marks]**.

W4ss Academy 2026

Section A

Question	Marks	Markscheme
1. (a)	**[2]**	evidence of finding $\frac{\sum x}{n}$ ***(M1)*** *eg* $\frac{0.3 + 0.4 + 3 + \ \ldots\ + 10}{10}$, $\frac{48.2}{10}$ $\bar{x} = 4.82$ (exact) ***A1 N2***
1. (b)	**[1]**	$p = 4.25$ (exact) ***A1 N1***
1. (c)	**[2]**	valid approach ***(M1)*** *eg* $Q_3 - Q_1$ $3 - 8$, 3 to 8 IQR = 5 ***A1 N2***
2. (a)	**[1]**	0.28 (s) ***A1***
2. (b)	**[3]**	IQR = 0.35 - 0.27 (= 0.08) (s) ***(A1)*** substituting **their** IQR into correct expression for upper fence ***(A1)*** $0.35 + 1.5 \times 0.08$ (= 0.47) (s) $0.46 < 0.47$ ***R1*** so 0.46 (s) is not an outlier ***AG***
2. (c)	**[1]**	**EITHER** the median is closer to the lower quartile (positively skewed) ***R1*** **OR** The distribution is positively skewed ***R1*** **OR** the range of reaction times below the median is smaller than the range of reaction times above the median ***R1*** **Note:** These are sample answers from a range of acceptable correct answers. Award ***R1*** for any correct statement that explains this. Do not award ***R1*** if there is also an incorrect statement, even if another statement in the answer is correct. Accept a correctly and clearly labelled diagram.

2. (d)	**[1]**	**EITHER** the distribution for 'not sleeping well' is centred at a higher reaction time ***R1*** **OR** The median reaction time after not sleeping well is equal to the upper quartile reaction time after sleeping well ***R1*** **OR** 75% of reaction times are < 0.35 seconds after sleeping well, compared with 50% after not sleeping well ***R1*** **OR** the sample size of 9 is too small to draw any conclusions ***R1*** **Note:** These are sample answers from a range of acceptable correct answers. Accept any relevant correct statement **that relates to the median and/or quartiles shown in the box plots**. **Do not accept** a comparison of means. Do not award ***R1*** if there is also an incorrect statement, even if another statement in the answer is correct. Award ***R0*** to "correlation does not imply causation".
3. (a)	**[2]**	1.01206 … , 2.45230 … $a = 1.01$, $b = 2.45$ $(1.01x + 2.45)$ ***A1A1***
3. (b)	**[1]**	0.981464 … $r = 0.981$ ***A1*** **Note:** A common error is to enter the data incorrectly into the GDC, and obtain the answers $a = 1.01700$ … , $b = 2.09814$ … and $r = 0.980888$ … Some candidates may write the 3 sf answers, ie. $a = 1.02$, $b = 2.10$ and $r = 0.981$ or 2 sf answers, ie. $a = 1.0$, $b = 2.1$ and $r = 0.98$. In these cases award ***A0A0*** for part (a) and ***A0*** for part (b). Even though some values round to an accepted answer, they come from incorrect working.
3. (c)	**[2]**	correct substitution of 78 into **their** regression equation ***(M1)*** 81.3930 … 81.23 from 3 sf answer 81 ***A1***

4.	**[4]**	**METHOD 1** **EITHER** one of P(A) = x + 0.16 OR P(B) = x + 0.36 ***A1*** **OR**  ***A1*** **THEN** attempt to equate their P($A \cap B$) with their expression for P(A) × P(B) ***M1*** P($A \cap B$) = P(A) × P(B) ⇒ x = (x + 0.16) × (x + 0.36) ***A1*** x = 0.24 ***A1*** **METHOD 2** attempt to form at least one equation in P(A) and P(B) using independence ***M1*** (P($A \cap B'$) = P(A) × $P(B')$ ⇒) P(A) × (1 - P(B)) = 0.16 OR (P($A' \cap B$) = P(A') × $P(B)$ ⇒) (1 - P(A)) × P(B) = 0.36 P(A) = 0.4 and P(B) = 0.6 ***A1*** P($A \cap B$) = P(A) × P(B) = 0.4 × 0.6 ***(A1)*** x = 0.24 ***A1***
5.	**[6]**	P($A \cup B$) = P(A) + P(B) - P($A \cap B$) = 0.68 substitution of P(A) · P(B) for P($A \cap B$) in P($A \cup B$) ***(M1)*** P(A) + P(B) - P(A)P(B) (= 0.68) substitution of 3P(B) for P(A) ***(M1)*** 3P(B) + P(B) - 3P(B)P(B) = 0.68 (or equivalent) ***(A1)*** **Note:** The first two ***M*** marks are independent of each other. attempts to solve their quadratic equation ***(M1)*** P(B) = 0.2, 1.133 … ($\frac{1}{5}$, $\frac{17}{15}$) P(B) = 0.2 (= $\frac{1}{5}$) ***A2*** **Note:** Award ***A1*** if both answers are given as final answers for P(B).